the correction for the schemes with a central O update can be expressed as

$$f_{2,[\![\text{ABOBA}]\!]} = f_{2,[\![\text{OBABO}]\!]} - \frac{\beta}{8}\left(|\nabla U(\boldsymbol{q})|^2 - \boldsymbol{p}^T\nabla^2 U\boldsymbol{p}\right), \tag{7.30a}$$

$$f_{2,[\![\text{BAOAB}]\!]} = f_{2,[\![\text{OABAO}]\!]} + \frac{\beta}{8}\left(|\nabla U(\boldsymbol{q})|^2 - \boldsymbol{p}^T\nabla^2 U\boldsymbol{p}\right). \tag{7.30b}$$

Thus we need only compute the solutions to the simpler PDE (7.29), in order to relate the solutions of the conjugate schemes with centralized OU update. However, although this simplifies our task greatly, finding the general solution to (7.29) is still very challenging, and obtaining a closed form expression for the correction function is nontrivial.

7.9.2 Expansion in Powers of γ^{-1}

Although we do not give the general analytical solution to the partial differential equation of (7.28) for each of the four symmetric Langevin schemes we are considering, we can easily find a solution in an important limiting case: the high friction regime. We consider expanding f_2 into a series involving the reciprocal friction $\varepsilon = 1/\gamma$, and assume that for large friction our ansatz remains sensible. We will denote

$$f_2(\boldsymbol{q},\boldsymbol{p}) = f_{2,0}(\boldsymbol{q},\boldsymbol{p}) + \varepsilon f_{2,1}(\boldsymbol{q},\boldsymbol{p}) + \varepsilon^2 f_{2,2}(\boldsymbol{q},\boldsymbol{p}) + \ldots \tag{7.31}$$

with each $\langle f_{2,i}\rangle = 0$. Dividing by the friction in (7.28) we obtain

$$\left(\bar{\mathcal{L}}_{\text{O}}^{\dagger} + \varepsilon\mathcal{L}_{H}^{\dagger}\right)(f_2\rho_\beta) = -\varepsilon\mathcal{L}_2^{\dagger}\rho_\beta,$$

where $\bar{\mathcal{L}}_{\text{O}}^{\dagger}$ is independent of the friction

$$\bar{\mathcal{L}}_{\text{O}}^{\dagger}\vartheta = \varepsilon\mathcal{L}_{\text{O}}^{\dagger}\vartheta = \nabla_p\cdot(\boldsymbol{p}\vartheta) + \frac{1}{\beta}\Delta_p\vartheta.$$

From our earlier computation in (7.29), we can write

$$\mathcal{L}_{2,[\![\text{OABAO}]\!]}^{\dagger}\rho_\beta = w(\boldsymbol{q},\boldsymbol{p})\rho_\beta,$$

for $w(\boldsymbol{q},\boldsymbol{p})$ independent of the friction

$$w(\boldsymbol{q},\boldsymbol{p}) = -\frac{\beta}{24}\left(6\boldsymbol{p}^T\left(\nabla^2 U\right)\nabla U - \nabla^3 U : \boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p}\right).$$

Hence we have

$$\left(\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}+\varepsilon\mathcal{L}_{H}^{\dagger}\right)\left(\rho_{\beta}(f_{2,0,[\![\mathrm{OABAO}]\!]}(\boldsymbol{q},\boldsymbol{p})+\varepsilon f_{2,1,[\![\mathrm{OABAO}]\!]}(\boldsymbol{q},\boldsymbol{p})+\ldots)\right)=-\varepsilon w(\boldsymbol{q},\boldsymbol{p})\rho_{\beta},$$

and equating powers of the reciprocal friction ε gives us a sequence of equations to solve

$$\begin{aligned}
&\text{Order 1:} && \bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta}f_{2,0,[\![\mathrm{OABAO}]\!]}\right)=0,\\
&\text{Order } \varepsilon\text{:} && \mathcal{L}_{H}^{\dagger}\left(\rho_{\beta}f_{2,0,[\![\mathrm{OABAO}]\!]}\right)+\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta}f_{2,1,[\![\mathrm{OABAO}]\!]}\right)=-w(\boldsymbol{q},\boldsymbol{p})\rho_{\beta},\\
&\text{Order } \varepsilon^{2}\text{:} && \mathcal{L}_{H}^{\dagger}\left(\rho_{\beta}f_{2,1,[\![\mathrm{OABAO}]\!]}\right)+\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta}f_{2,2,[\![\mathrm{OABAO}]\!]}\right)=0,\\
&\text{Order } \varepsilon^{3}\text{:} && \mathcal{L}_{H}^{\dagger}\left(\rho_{\beta}f_{2,2,[\![\mathrm{OABAO}]\!]}\right)+\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta}f_{2,3,[\![\mathrm{OABAO}]\!]}\right)=0,\\
&&&\vdots
\end{aligned}$$

The first equation at order one tells us that $f_{2,0,[\![\mathrm{OABAO}]\!]}(\boldsymbol{q},\boldsymbol{p})$ must be a function of position only, as the kernel of $\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}$ is the set of functions that are constant with respect to p. Hence we shall denote

$$f_{2,0,[\![\mathrm{OABAO}]\!]}(\boldsymbol{q},\boldsymbol{p})=\nu(\boldsymbol{q}),$$

where we our goal is now to explicitly find $\nu(\boldsymbol{q})$. Unfortunately, $\nu(\boldsymbol{q})$ is not explicitly determined at this point, we only know that it is a function of position only such that $\langle\nu\rangle=0$. This function will be determined as we solve the higher order equations.

In order to determine $\nu(\boldsymbol{q})$, we will have to solve the next PDE at order ε,

$$\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta}f_{2,1,[\![\mathrm{OABAO}]\!]}\right)+\mathcal{L}_{H}^{\dagger}\left(\rho_{\beta}f_{2,0,[\![\mathrm{OABAO}]\!]}\right)=-w(\boldsymbol{q},\boldsymbol{p})\rho_{\beta},\tag{7.32}$$

where we will make use of the formulas

$$\begin{aligned}
\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta}\nabla^{3}U:\boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p}\right)&=-3\rho_{\beta}\nabla^{3}U:\boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p}+\frac{6\rho_{\beta}}{\beta}\boldsymbol{p}^{T}\nabla_{q}\Delta_{q}U,\\
\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta}\boldsymbol{p}^{T}\nabla_{q}\Delta_{q}U\right)&=-\rho_{\beta}\boldsymbol{p}^{T}\nabla_{q}\Delta_{q}U,\\
\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\rho_{\beta}\boldsymbol{p}^{T}\left(\nabla^{2}U\right)\nabla U\right)&=-\rho_{\beta}\boldsymbol{p}^{T}\left(\nabla^{2}U\right)\nabla U,
\end{aligned}$$

to write

$$w(\boldsymbol{q},\boldsymbol{p})\rho_{\beta}=\frac{\beta}{24}\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\left[6\boldsymbol{p}^{T}\left(\nabla^{2}U\right)\nabla U-\frac{1}{3}\nabla^{3}U:\boldsymbol{p}\otimes\boldsymbol{p}\otimes\boldsymbol{p}-\frac{2}{\beta}\boldsymbol{p}^{T}\nabla_{q}\Delta_{q}U\right]\rho_{\beta}\right).\tag{7.33}$$

Additionally, for $\nu(\boldsymbol{q})$ we have the identity

$$\mathcal{L}_{H}^{\dagger}\left(\rho_{\beta}\nu\right)=-\boldsymbol{p}^{T}\nabla\nu\rho_{\beta}=\bar{\mathcal{L}}_{\mathrm{O}}^{\dagger}\left(\left[\boldsymbol{p}^{T}\nabla\nu+\mu(\boldsymbol{q})\right]\rho_{\beta}\right),\tag{7.34}$$